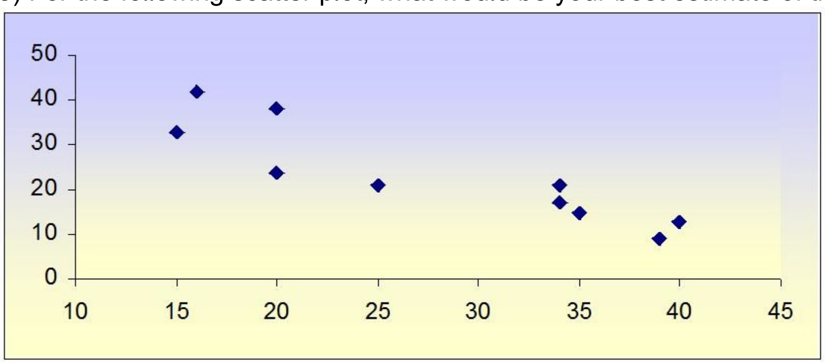
Statistical Methods for Computer Science

Exercises: Describing Data: Numerical

1) If you are interested in comparing variation in sales for small and large stores selling similar goods, which of the following is the most appropriate measure of dispersion?
A) the range
B) the interquartile range
C) the standard deviation
D) the coefficient of variation

2) Suppose you are told that the mean of a sample is below the median. What does this information suggest about the distribution?
A) The distribution is symmetric.
B) The distribution is skewed to the right or positively skewed.
C) The distribution is skewed to the left or negatively skewed.
D) There is insufficient information to determine the shape of the distribution.

3) For the following scatter plot, what would be your best estimate of the correlation coefficient?



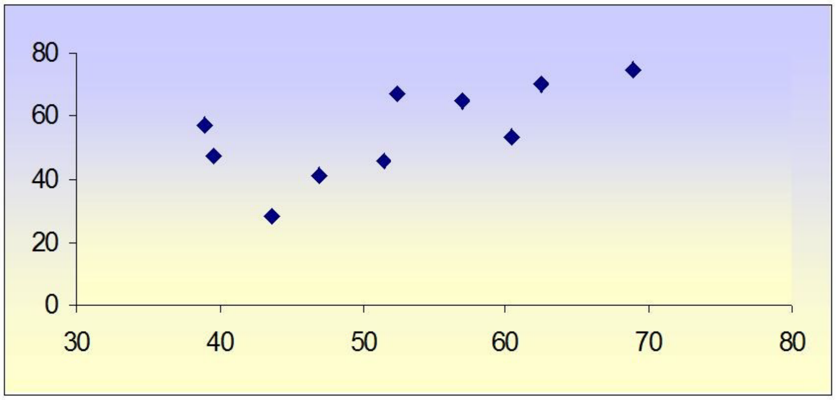
A) -0.8
B) -1.0
C) 0.0
D) -0.3

4) Given a set of 25 observations, for what value of the correlation coefficient would we be able to say that there is evidence that a relationship exists between the two variables?
A) $|r_{xy}| \geq 0.40$
B) $|r_{xy}| \geq 0.35$
C) $|r_{xy}| \geq 0.30$
D) $|r_{xy}| \geq 0.25$

5) Which of the following statements is true about the correlation coefficient and covariance?
A) The covariance is the preferred measure of the relationship between two variables since it is generally larger than the correlation coefficient.
B) The correlation coefficient is a preferred measure of the relationship between two variables since its calculation is easier than the covariance.
C) The covariance is a standardized measure of the linear relationship between two variables.

D) The covariance and corresponding correlation coefficient are represented by different signs, one is negative while the other is positive and vice versa.

6) For the following scatter plot, what would be your best estimate of the correlation coefficient?



A) 1.0
B) 0.7
C) 0.3
D) 0.1

7) Which of the following descriptive statistics is least affected by outliers?
A) mean
B) median
C) range
D) standard deviation

8) Which of the following statements is true?
A) The correlation coefficient is always greater than the covariance.
B) The covariance is always greater than the correlation coefficient.
C) The covariance may be equal to the correlation coefficient.
D) Neither the covariance nor the correlation coefficient can be equal to zero.

9) Which measures of central location are not affected by extremely small or extremely large data values?
A) arithmetic mean and median
B) median and mode
C) mode and arithmetic mean

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

A recent survey asked respondents about their monthly purchases of raffle tickets. The monthly expenditures, in dollars, of ten people who play the raffle are 23, 15, 11, 20, 28, 35, 13, 10, 20, and 24.

21) What can we say about the shape of the distribution of monthly purchases of raffle tickets?
A) Skewed to the left
B) Skewed to the right
C) Approximately bell-shaped
D) None of the above

22) Which of the following statements is not true?
A) The 75^{th} percentile is equal to 23.5.
B) The median is equal to the mode.
C) The mean is 19.9.
D) The distribution is approximately symmetric.

23) Over the past 10 years, the return on Stock A has averaged 8.4% with a standard deviation of 2.1%. The return on Stock B has averaged 3.6% with a standard deviation of 0.9%. Which of the following statements is true?
A) Stock A has smaller relative variation than Stock B.
B) Stock B has smaller relative variation than Stock A.
C) Both stocks exhibit the same relative variation.
D) Unable to tell with the given information.

24) The median value of the data values 12, 32, 48, 8, 22, 9, 30, and 18 equals:
A) 20
B) 22
C) 24
D) 26

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

The police lieutenant in charge of the traffic division reviews the number of traffic citations issued by each of the police officers in his division. He finds that the mean number of citations written by each officer is 23.2 citations per day, with a standard deviation of 3.1. Assume that the distribution of the number of tickets issued is approximately bell-shaped.

26) The coefficient of variation for the number of citations is:
A) 13.36%
B) 7.48%
C) 6.68
D) Cannot be determined without the sample size

27) Suppose that you are also told that the median for these data was 19.3. Which of the following statements is true about the shape of the distribution?
A) It is skewed to the right.
B) It is skewed to the left.
C) It is approximately symmetric.
D) Cannot be determined without more information

30) What is the relationship among the mean, median, and mode in a symmetrical distribution?
A) They are all equal.
B) The mean is always the smallest value.
C) The mean is always the largest value.
D) The mode is the largest value.

TRUE/FALSE QUESTIONS

102) The sample covariance must take a value between -1 and +1 inclusive.

103) The sample covariance may never be negative.

104) The correlation coefficient measures the strength of a linear relationship between two variables.

105) A correlation coefficient of zero indicates a lack of relationship between the two variables of interest.

106) The value of the correlation coefficient may be used to confirm a non-linear relationship.

107) The mean is generally the preferred measure of central tendency to describe numerical data, but not categorical data.

108) For any set of numerical data values arranged in an ascending or descending order, the value of the observation in the center is called the weighted mean.

109) Categorical data are best described by the mode.

110) The median should always be preferred to the mean when the population or sample is skewed to the right or left.

111) One possible source of skewness is the presence of outliers, and sometimes skewness is simply inherent in the distribution.

112) Although the range measures the total spread of the data, the interquartile range (IQR) measures only the spread of the middle 50% of the data.

113) In a negatively skewed distribution, the mean is always greater than the median.

114) The coefficient of variation measures variability in a positively skewed data set relative to the size of the median.

115) When the data values are arranged in an ascending order, the third quartile (Q_3) is located at the $0.75(\mathrm{n} + 1)^{\mathrm{th}}$ position, and first quartile (Q_1) is located at the $0.25(n + 1)^{\mathrm{th}}$ position.

116) The five-number summary refers to the five descriptive measures: minimum, mean, median, mode, and maximum; therefore it is sometimes known as the five-m summary.

117) If the interquartile range for a set of data is 10 minutes, this means that the data have a spread of only 10 minutes.

118) Since the interquartile range takes into account only two of the data values, it is susceptible to considerable distortion if there are unusual numbers of extreme observations (outliers).

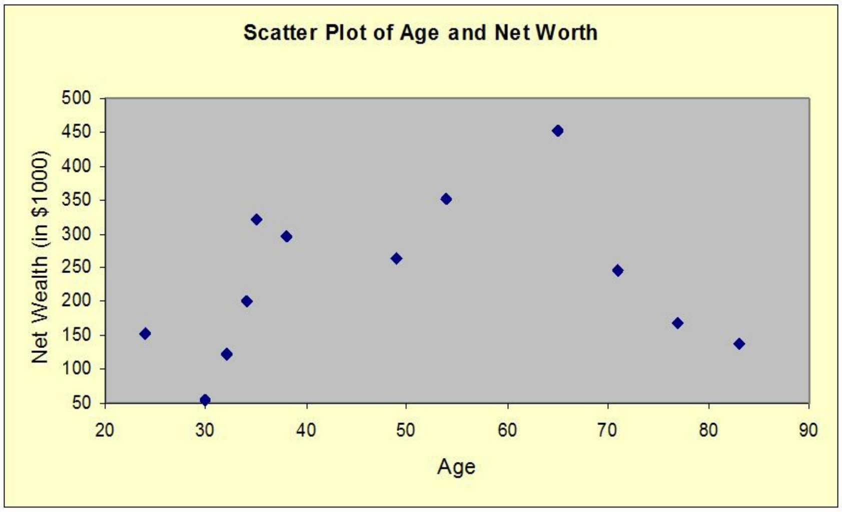
119) Although the range and interquartile range measure the spread of data, both measures take into account only two of the data values, regardless of the size of the data.

120) For any symmetrical distribution, the standard deviation is equal to the variance.

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

A researcher is interested in examining how the net wealth of individuals changes over the course of their lifetimes. She has collected the following data regarding the age *X*, in years, and net worth *Y*, measured in thousands of dollars, of 12 individuals in the form of (*x*, *y*) pairs: (24, 153), (34, 201), (38, 297), (83, 139), (77, 167), (32, 123), (71, 247), (49, 263), (54, 352), (35, 321), (65, 453), and (30, 54).

199) Prepare a scatter plot of this data.



200) What conclusions can you draw about the relationship between age and net wealth?

201) Calculate the correlation coefficient between the age and net worth of individuals.

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

The police lieutenant in charge of the traffic division has reviewed the number of traffic citations issued per day by each of the 10 police officers in his division. The data were: 13, 21, 12, 34, 31, 13, 22, 26, 25, and 23.

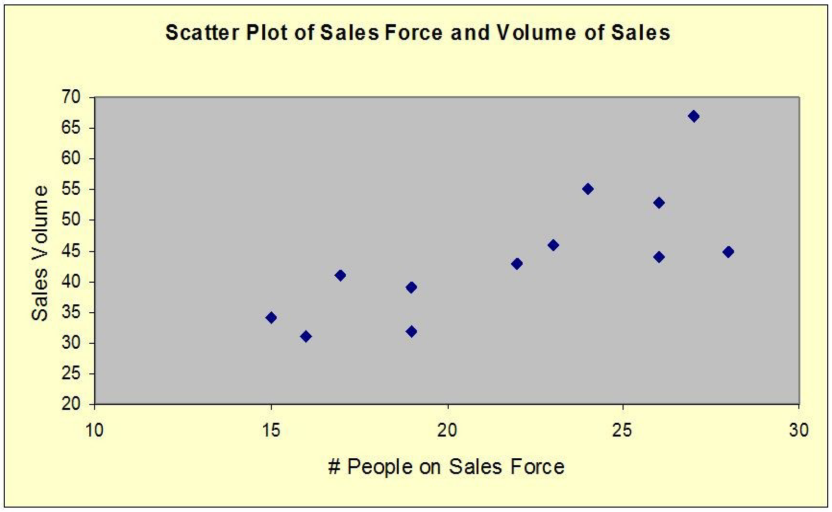
203) What is the standard deviation for the number of citations issued per day?

204) What is the interquartile range for the number of citations issued per day?

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

You are interested in looking at the relationship between the number of people on the sales force and the dollar volume of sales. The following data show gross sales, *Y*, measured in millions of dollars, and the number of people on the sales force, *X*, in the form of (*x*, *y*) pairs for 12 people: (15, 34), (24, 55), (27, 67), (16, 31), (19, 32), (26, 44), (19, 39), (23, 46), (26, 53), (22, 43), (28, 45), and (17, 41).

205) Prepare a scatter plot of this data.

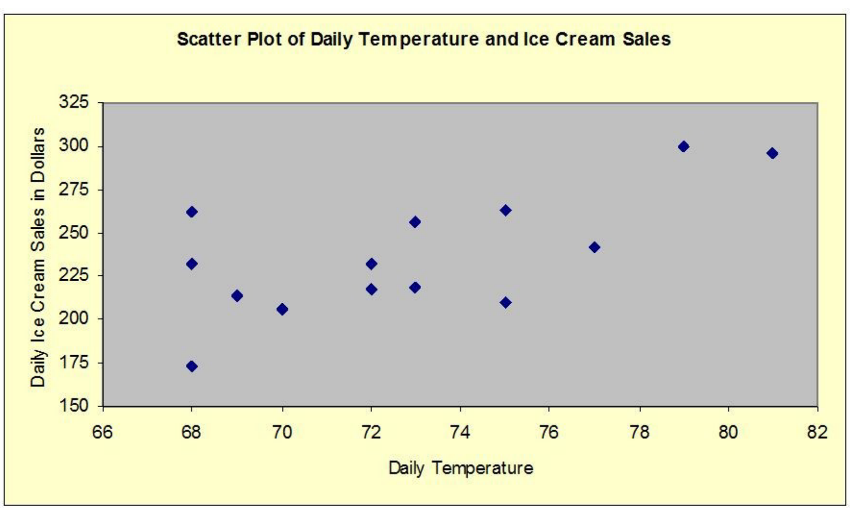


206) What conclusions can you draw about the relationship between the size of the sales force and gross sales?

207) Calculate the correlation coefficient between the size of the sales force and gross sales.

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

Ahmed is trying to figure out the relationship between daily sales from his ice cream truck and the daily temperature in Dearborn, Michigan during the month of August. He has collected data on sales *Y*, measured in dollars, and temperature *X*, measured in Fahrenheit degrees, over the past 14 days in the form of (*x*, *y*) pairs: (72, 232), (77, 242), (73, 219), (69, 214), (68, 262), (72, 218), (75, 263), (70, 206), (79, 300), (73, 256), (68, 173), (75, 210), (81, 296), and (68, 232).



209) Prepare a scatter plot of this data.

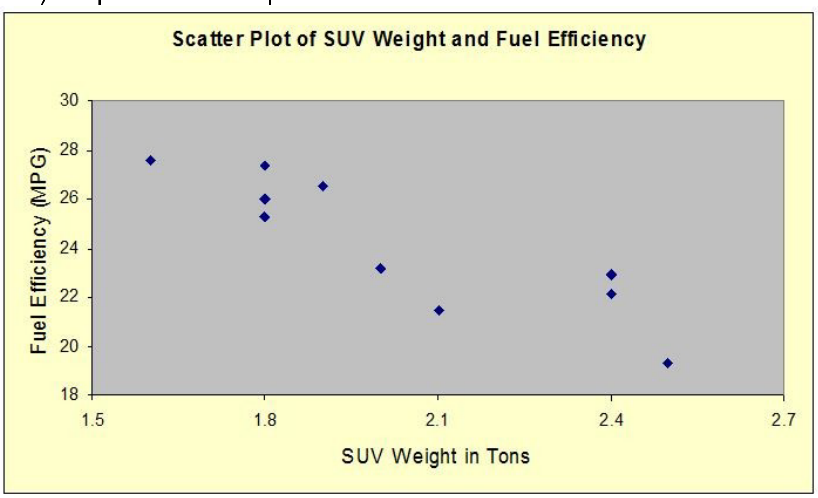
210) What conclusions can you draw about the relationship between ice cream sales and daily temperature?

211) Calculate the correlation coefficient between ice cream sales and temperature.

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

Some people would argue that the trend of consumers toward purchasing large sport utility vehicles (SUVs) has been detrimental to the environment. The following data show the vehicle weight *X*, measured in tons, and the corresponding fuel efficiency *Y*, measured by the average miles per gallon (mpg), in the form of (*x*,*y*) pairs for 10 vehicles: (1.9, 26.5), (2, 23.2), (2.4, 22.1), (1.8, 26), (2.4, 22.9), (2.1, 21.5), (1.8, 25.3), (2.5, 19.3), (1.8, 27.4), and (1.6, 27.6).

219) Prepare a scatter plot of this data.



220) What conclusions can you draw about the relationship between weight and fuel efficiency?

221) Calculate the correlation coefficient between vehicle weight and fuel efficiency.

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

The annual percentage returns on two stocks over a 7-year period were as follows:

Stock A:	4.01%	14.31%	19.01%	-14.69%	-26.49%	8.01%	5.81%
Stock B:	6.51%	4.41%	3.81%	6.91%	8.01%	5.81%	5.11%

226) Compare the means of these two population distribution.

227) Compare the standard deviations of these two population distributions.

228) Compute an appropriate measure of dispersion for both stocks to measure the risk of these investment opportunities. Which stock is more volatile?

229) Calculate the coefficient of variation for the following sample data: 13.2, 14.7, 17.2, 12.1, 21.8, 8.4, 14.3, 11.0, 9.3, and 8.7

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

In a recent survey, 12 students at a local university were asked approximately how many hours per week they spend on the Internet. Their responses were: 13, 0, 5, 8, 22, 7, 3, 0, 15, 12, 13, and 17.

232) What are the mean and standard deviation for this data?

233) What is the coefficient of variation for this data?

234) From the data presented above, calculate the interquartile range.

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

In a recent survey, 200 top executives were asked how many hours they spend each year in community service. The data are presented below.

# of Hours	0 but < 20	20 but < 40	40 but < 60	60 but < 80	80 but < 100	100 but < 120	120 but < 140
# of Executives	11	27	33	53	47	22	7

235) Calculate the quantities $\sum f_i$, $\sum f_i m_i$, and $\sum f_i (m_i - \overline{x})^2$.

236) What is the estimated mean amount of time spent by these executives in community service?

237) What is the estimated standard deviation for this data?

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

The number of students eating breakfast at the school dining commons was recorded over 110 days last semester. These data are presented below.

# of Students	160 but < 190	190 but < 220	220 but < 250	250 but < 280	280 but < 310
# of Days	11	27	42	23	7

238) Calculate the quantities $\sum f_i$, $\sum f_i m_i$, and $\sum f_i (m_i - \bar{x})^2$.

239) What is the estimated mean number of students showing up for breakfast?

240) What is the estimated standard deviation for this data?

USE THIS INFORMATION TO ANSWER THE NEXT QUESTIONS

A small accounting office is trying to determine its staffing needs for the coming tax season. The manager has collected the following data: 46, 27, 79, 57, 99, 75, 48, 89, and 85. These values represent the number of returns the office completed each year over the entire nine years it has been doing tax returns.

241) For this data, what is the mean number of tax returns completed each year?

242) For this data, what is the median number of tax returns completed each year?

243) For this data, what is the variance of the number of tax returns completed each year?

244) For this data, what is the interquartile for the number of tax returns completed each year?

245) For this data, what is the coefficient of variation for the number of tax returns completed each year?

246) Compute the mean, standard deviation and interquartile range for the following sample data: 12.2, 15.9, 8.1, 19.2, 13.7, 7.2, 12.2, 10.9, 5.4, and 16.8.

247) The manager of 45 sales people examined their monthly expenditures on entertaining clients. He found that the mean amount was $237.50 with a standard deviation of $27.40. Assuming the data is bell-shaped, would a claim for the amount of $300 be considered unlikely? Why or why not?

248) A researcher interested in determining the average monthly expenditures of college students on DVDs finds that for a sample of 25 students, the mean expenditure was $24.40, and the median expenditure was $21.76. Specify the shape of the histogram for this data. Does this shape make sense? Why?

249) The following values represent the annual snowfall, measured in inches, at a local ski resort over the past 10 years: 123, 87, 143, 133, 182, 176, 96, 104, 201, and 152. What is the mean snowfall over the past 10 years? What is the standard deviation?